JONGIN CHOI

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EDUCATION

Pennsylvania State University ***May 2025***
Bachelor of Science, Applied Data Sciences *University Park, PA*

- Coursework: Data Analysis, Software Engineering, Statistics, Algorithms, Artificial Intelligence, Machine Learning

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, TypeScript
Data & ML: Pandas, NumPy, Scikit-learn, PyTorch, HuggingFace Transformers, Jupyter, Spark
Web & Scraping: Playwright, Beautiful Soup, HTML/CSS
Databases & Tools: MySQL, MongoDB, Git, Linux, Docker

EXPERIENCE

Research Assistant — Multimodal Emotion Dataset (FeelReel) ***Jun 2024 – Mar 2025***
Pennsylvania State University, Data Science Lab (Prof. Sharon Hwang) · IST 495 internship credit *University Park, PA*

- Supported the follow-up phase of FeelReel, a lab benchmark of 400K+ Reddit images and 5M+ comments for emotion understanding (earlier version submitted to CVPR 2024).
- Sourced and organized Reddit image posts and comments through the project's data collection pipeline, keeping datasets structured and reproducible.
- Built the MTurk survey (HIT) page and collected human evaluations of the dataset's emotion annotations.

PROJECTS

FeelReel-Mini — Personal recreation ***Jul – Oct 2026***
github.com/nopgae/feelreel-mini *PyTorch, Hugging Face, PRAW, LLaVA (Ollama), matplotlib*

- Rebuilt the FeelReel data pipeline at small scale with public tools: Reddit image posts and comments, GoEmotions pseudo-labels (RoBERTa, 28 emotions), and LLaVA 7B asked what in each image triggered the emotion.
- Charted emotion distributions across 8 image-heavy subreddits: scenery and woodworking communities leaned toward admiration and surprise, pet communities toward joy.
- Separate from the lab's code and data, which are not public.

Medical Transcription Classifier — Penn State DS 310 ***Fall 2024***

- Classified medical transcriptions into 6 specialties with TF-IDF (1–3 grams) and document-length features; weighted F1 0.807.
- Tuned XGBoost over 54 configurations with 5-fold CV (270 fits), then showed untuned logistic regression still won (F1 0.763 vs 0.711) and traced it to sparse features.

Reddit Sentiment Pipeline — Personal ***Aug 2025 – Mar 2026***
github.com/nopgae/reddit-sentiment-roberta *PyTorch, Hugging Face, pandas, Plotly*

- Built an end-to-end pipeline alone: Reddit JSON API collection, batched inference with a pre-trained RoBERTa sentiment model, aggregation and charts.
- Scored 563 posts across 7 e-commerce, logistics and supply-chain subreddits; quantified community baselines (r/supplychain +0.11 vs r/ecommerce –0.31).

Starbucks Korea Parking Finder — Personal ***Apr 2026 (updates Oct–Nov 2026)***
github.com/nopgae/starbucks-parking-finder *TypeScript, Playwright, Python, GitHub Actions*

- Used Playwright to find the JSON request behind the store locator, then called it directly to collect parking notes for all 2,212 stores (2,199 with a note) instead of checking each store page by hand.
- Wrote a rule-based parser that turns free-text notes into free / free-with-purchase / paid / unavailable; accuracy on a 50-note sample rose from 84% to 94%, with 21 unit tests run on every push by GitHub Actions.
- Packaged per-region data files and a search script as a skill so AI assistants answer from each store's own note.

ADDITIONAL INFORMATION

Languages: Fluent in English and Korean (bilingual); TOEIC Speaking 160 (AL)
Interests: Web security CTF (webhacking.kr), Boston Dynamics Spot robotics